Queen Mary
University of London

EBU5376
Digital Signal Processing

Finite Impulse Response (FIR) Filters

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Outline

- Definition of Finite Impulse Response (FIR) Filter
- How to Design a FIR Filter
- Window Method
- FIR Example: Moving Average Filter

There will be MATLAB exercises throughout this lecture.

Outline

- Definition of Finite Impulse Response (FIR) Filter
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Finite Impulse Response (FIR) Filter

- If the impulse response $h[n]$ can be written as a <u>finite sequence</u>, the filter is called a Finite Impulse Response (FIR) filter.
- This happens **when the difference equation does not contain past outputs** $y[n-1], y[n-2]$ ...

Difference equation of FIR filters:

$$y[n] = \sum_{j=0}^{M} b_j x[n-j]$$

Finite Impulse Response (FIR) Filter

- Consider the general transfer function formula

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}, \qquad a_0 = 1$$

- It is a **FIR filter** if and only if $a_k = 0, \forall k > 1$.

Transfer Function of FIR filter:

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{M} b_k z^{-k}$$

Impulse response of FIR filter:

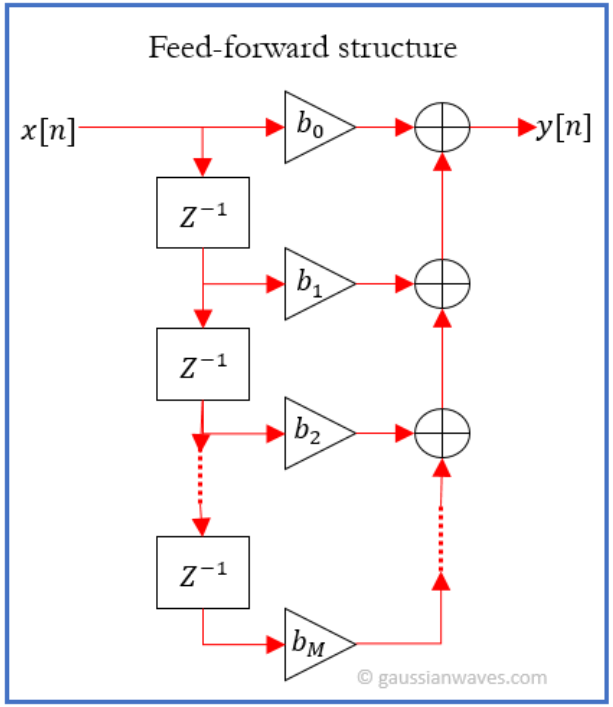
$$h[n] = \begin{cases} b_k & 0 \le k \le M \\ 0 & else \end{cases}$$

Finite Impulse Response (FIR) Filter

- Characteristics of FIR filters:
  - The filter output is independent of its past outputs
  - All the poles are at the origin. Hence the stability of FIR filters is guaranteed.
  - The impulse response is a finite sequence, hence FIR filters can be directly implemented by convolving the inputs with the impulse response
  - FIR filters usually have linear phase response

Finite Impulse Response (FIR) Filter

- Typical Block Diagram of FIR filters:



- It is a feed-forward structure.
- No past outputs are involved.

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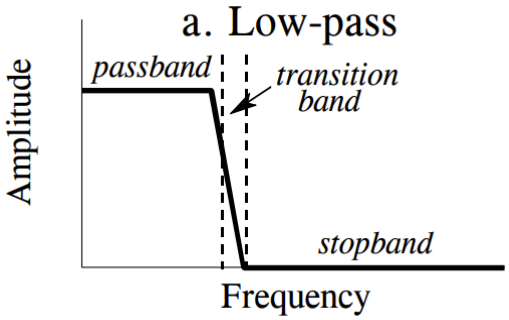
Finite Impulse Response (FIR) Filter

- To design a filter for a certain purpose (low-pass, high-pass,…), we need to determine the values of a_n and b_n in the difference equation / transfer function of the filter.
- For FIR filters, the transfer function is given by

$$H(z) = \sum_{k=0}^{M} b_k z^{-k}$$

- Hence, the design objective is to obtain the values of b_n such that the amplitude response shows the desired behavior.

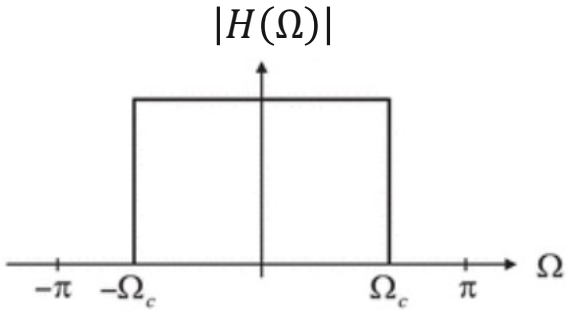
Finite Impulse Response (FIR) Filter



- Let's start with an ideal low-pass filter with a cut-off frequency Ω_c and no transition band

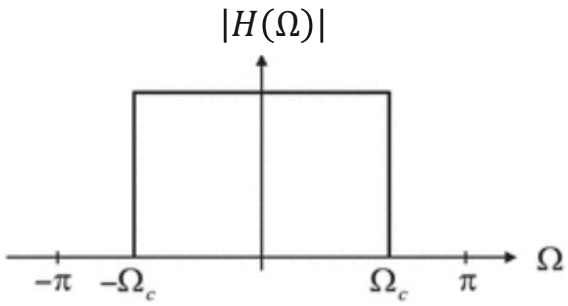
- The amplitude response of the ideal filter is

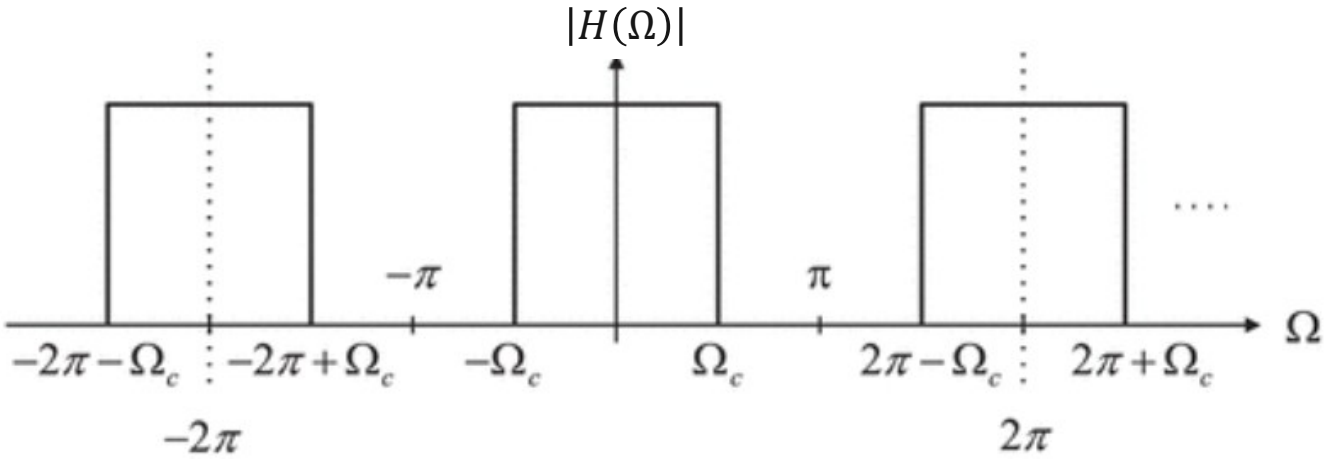
$$|H(\Omega)| = \begin{cases} 1, & 0 \le |\Omega| \le \Omega_c \\ 0, & \Omega_c \le |\Omega| \le \pi. \end{cases}$$



Finite Impulse Response (FIR) Filter

- Since the frequency response is the DTFT of the impulse response, it is in fact a periodic function
- The periodicity is illustrated as:



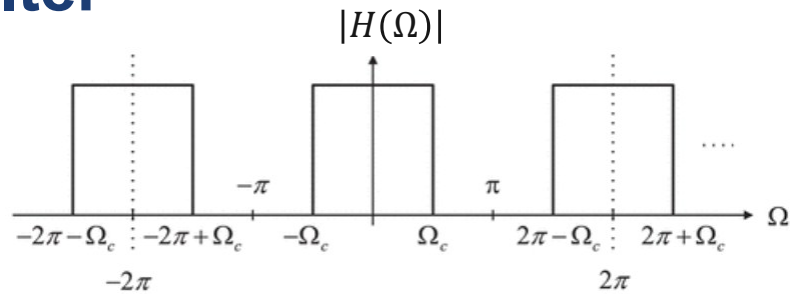


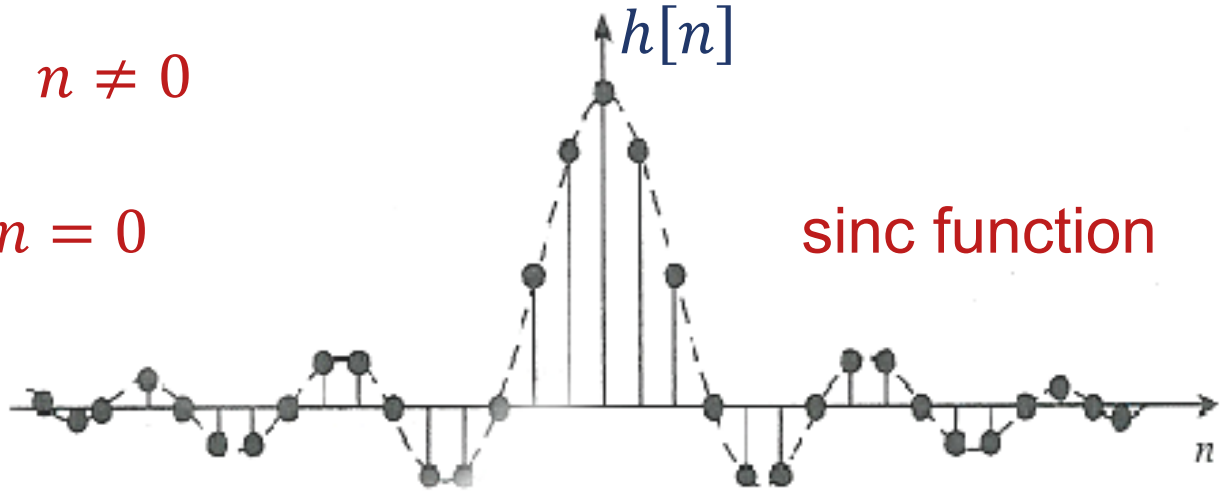
Finite Impulse Response (FIR) Filter

- The IDTFT of this box function is

$$h[n] = \frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{j\omega n} d\omega$$

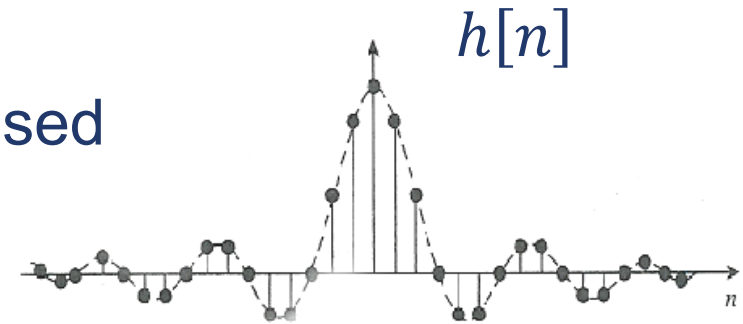
$$= \begin{cases} \dfrac{\sin \Omega_c n}{\pi n} & n \neq 0 \\ \dfrac{\Omega_c}{\pi} & n = 0 \end{cases}$$





sinc function

Finite Impulse Response (FIR) Filter

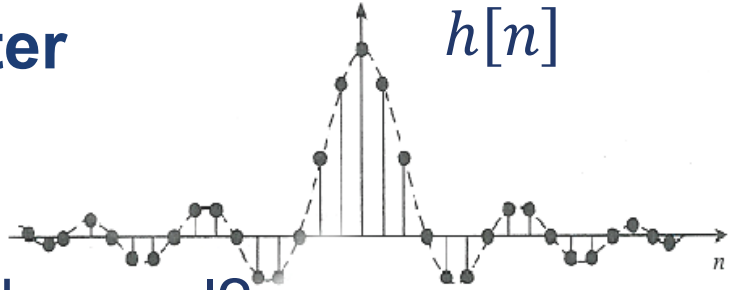


- This impulse response <u>cannot</u> be used to design FIR filters. WHY?

- *Reason 1*: It is an infinite sequence, but FIR filters must have a finite impulse response

- *Reason 2*: It is a non-causal sequence. There are non-zero impulse response for n<0, which is not practical.

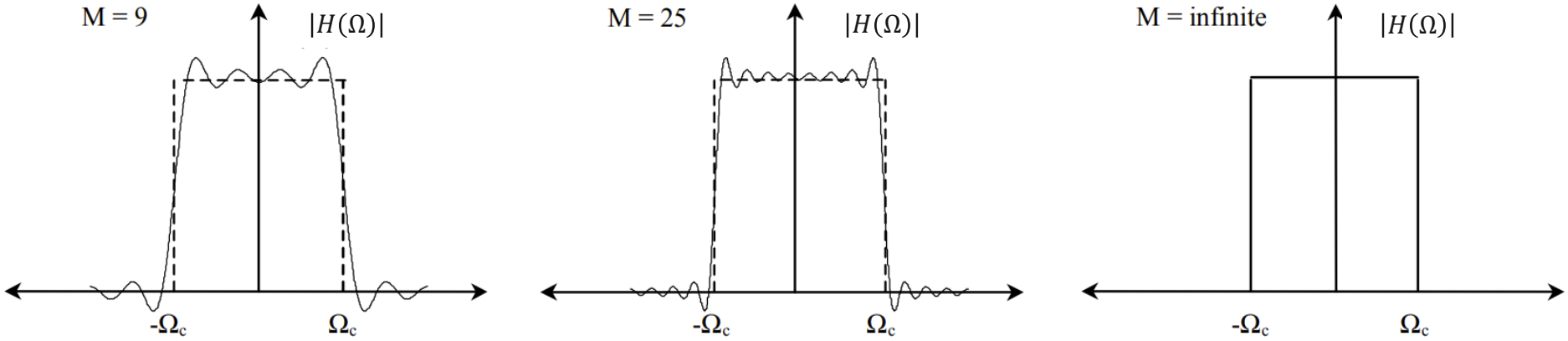
Finite Impulse Response (FIR) Filter



- How to make this sequence <u>finite</u> and <u>causal</u>?
  - *Solution 1:* Truncate → Finite
  - *Solution 2:* Shift response → Causal
- Unfortunately, there are *side-effects*
  - Truncation will introduce ripples and overshoots in amplitude response

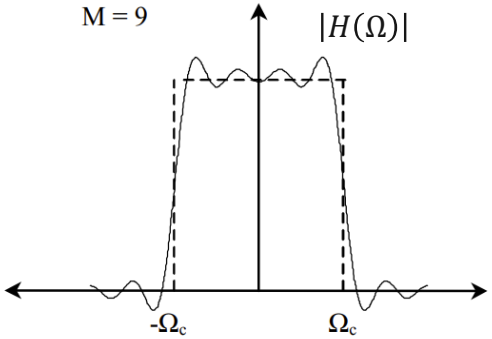
FIR Filter - Gibb's Phenomenon

- Figure shows effects on the amplitude response of truncating the ideal impulse response. This is called Gibb's Phenomenon



- M is the number of non-zeros in the impulse response
- The height of the ripples is not dependent on the value of M.
- M only affects the width.

Finite Impulse Response (FIR) Filter



- Possible solutions to combat ripples:
  - Reduce the amount of truncated values — Increase complexity and time delay
  - Use window functions — The preferred option

✓ The heights of the ripples are controlled by the type of window function used for the time-domain multiplication

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FIR Filter – Window Method

- A window function is a *symmetrical, discrete, finite sequence*
- Instead of setting the frequency response to zero outside a range, it is better to gradually reduce the values to zero.
- The window function is applied by multiplying it with the impulse response

$$h_w[n] = h[n] \cdot w[n]$$

Impulse response after windowing

Ideal impulse response

Elementwise multiplication

Window function

FIR Filter – Window Method

- Three common window functions:

The length of the filter will be $2M + 1$

- **Rectangular window**

$$w_{rec}[n] = 1, \qquad -M \leq n \leq M$$

- **Hamming window**

$$w_{ham}[n] = 0.54 + 0.46\cos\left(\frac{n\pi}{M}\right), \qquad -M \leq n \leq M$$

- **Hanning window**

$$w_{han}[n] = 0.5 + 0.5\cos\left(\frac{n\pi}{M}\right), \qquad -M \leq n \leq M$$

- *Equations will be provided in exams*

FIR Filter – Window Method

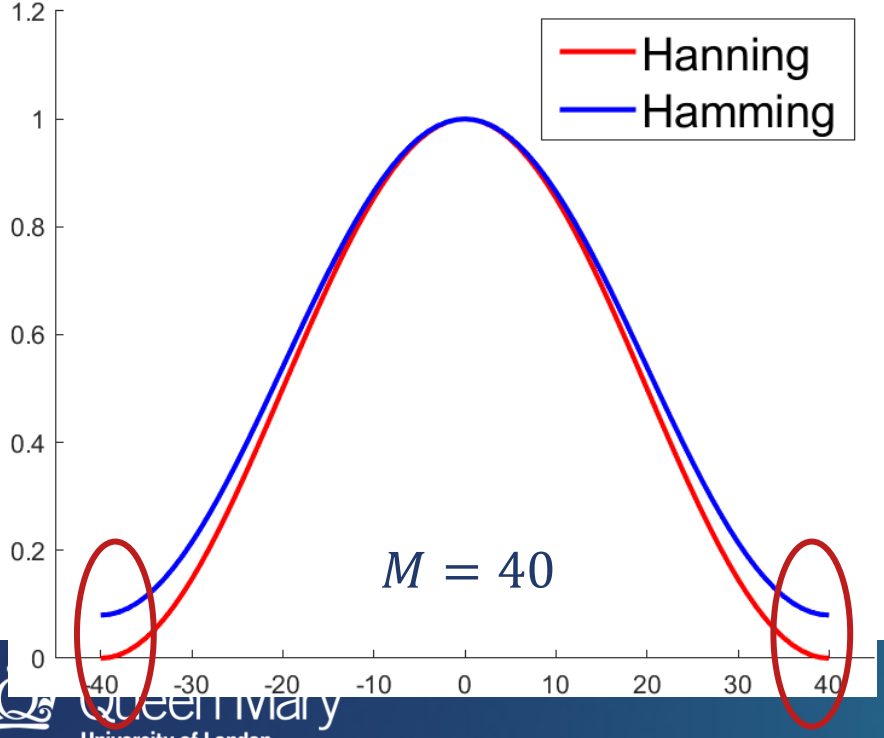
❑ Rectangular window

$$w_{rec}[n] = 1, \qquad -M \leq n \leq M$$

- ❖ values outside the range is set to zero
- ❖ values inside the range remain unchanged
- ✓ This is equivalent to abrupt truncation

FIR Filter – Window Method

- **Hamming window** $w_{ham}[n] = 0.54 + 0.46\cos\left(\frac{n\pi}{M}\right),\ -M \le n \le M$
- **Hanning window** $w_{han}[n] = 0.5 + 0.5\cos\left(\frac{n\pi}{M}\right),\ -M \le n \le M$

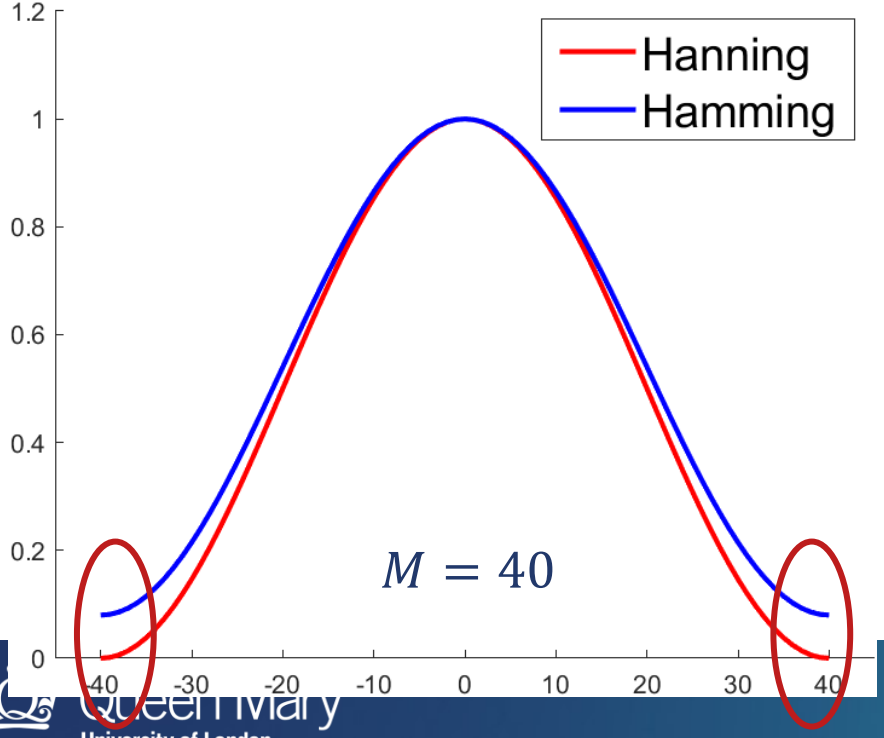


Shape:

- **Hamming window** has a more abrupt transition near the boundaries
- **Hanning window** has a smoother transition near the boundaries

FIR Filter – Window Method

- **Hamming window** $w_{ham}[n] = 0.54 + 0.46\cos\left(\frac{n\pi}{M}\right),\ -M \le n \le M$
- **Hanning window** $w_{han}[n] = 0.5 + 0.5\cos\left(\frac{n\pi}{M}\right),\ -M \le n \le M$



Usages:

- **Hamming window** is preferred for eliminating the high frequency noise
- **Hanning window** is preferred for smoothing

FIR Filter – Window Method

- <u>Example</u>: Given the filter impulse response

$$h[0] = 0.25, \qquad h[-1] = h[1] = 0.2251,$$

$$h[-2] = h[2] = 0.1592, \qquad h[-3] = h[3] = 0.0750$$

1. Apply the Hanning window function to obtain $h_{han}[n]$
2. Apply the Hamming window function to obtain $h_{ham}[n]$
3. Plot $h[n], h_{han}[n], h_{ham}[n]$

FIR Filter – Window Method

- Example: Given the filter impulse response

$h[0] = 0.25, \quad h[-1] = h[1] = 0.2251, \quad h[-2] = h[2] = 0.1592, \quad h[-3] = h[3] = 0.0750$

1. Apply the **Hanning window** function to obtain $h_{han}[n]$

First calculate the coefficients of the window function

$$w_{han}[n] = 0.5 + 0.5\cos\left(\frac{n\pi}{3}\right), \qquad -3 \le n \le 3$$

$w_{han}[0] = 1, \quad w_{han}[-1] = w_{han}[1] = 0.75, \quad w_{han}[-2] = w_{han}[2] = 0.25,$

$w_{han}[-3] = w_{han}[3] = 0$

Then multiply the window function with the filter

$h_{han}[0] = 0.25, \quad h_{han}[-1] = h_{han}[1] = 0.1688, \quad h_{han}[-2] = h_{han}[2] = 0.0398,$

$h_{han}[-3] = h_{han}[3] = 0$

FIR Filter – Window Method

- Example: Given the filter impulse response

$h[0] = 0.25, \quad h[-1] = h[1] = 0.2251, \quad h[-2] = h[2] = 0.1592, \quad h[-3] = h[3] = 0.0750$

2. Apply the **Hamming window** function to obtain $h_{ham}[n]$

First calculate the coefficients of the window function

$$w_{ham}[n] = 0.54 + 0.46\cos\left(\frac{n\pi}{3}\right), \qquad -3 \le n \le 3$$

$w_{ham}[0] = 1, \quad w_{ham}[-1] = w_{ham}[1] = 0.77,$

$w_{ham}[-2] = w_{ham}[2] = 0.31, \; w_{ham}[-3] = w_{ham}[3] = 0.08$

Then multiply the window function with the filter

$h_{ham}[0] = 0.25, \quad h_{ham}[-1] = h_{ham}[1] = 0.1733, \quad h_{ham}[-2] = h_{ham}[2] = 0.0494,$

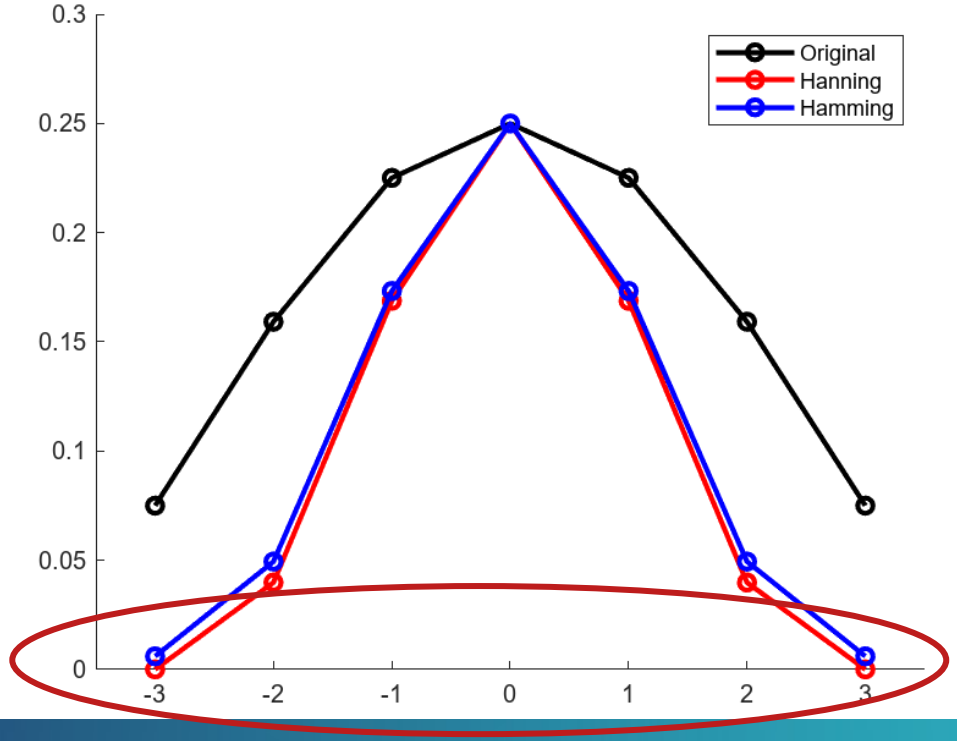
$h_{ham}[-3] = h_{ham}[3] = 0.006$

FIR Filter – Window Method

- Example: Given the filter impulse response

  3. Plot $h[n], h_{han}[n], h_{ham}[n]$

The filter is still non-causal.
We need to shift the
sequences to the positive axis

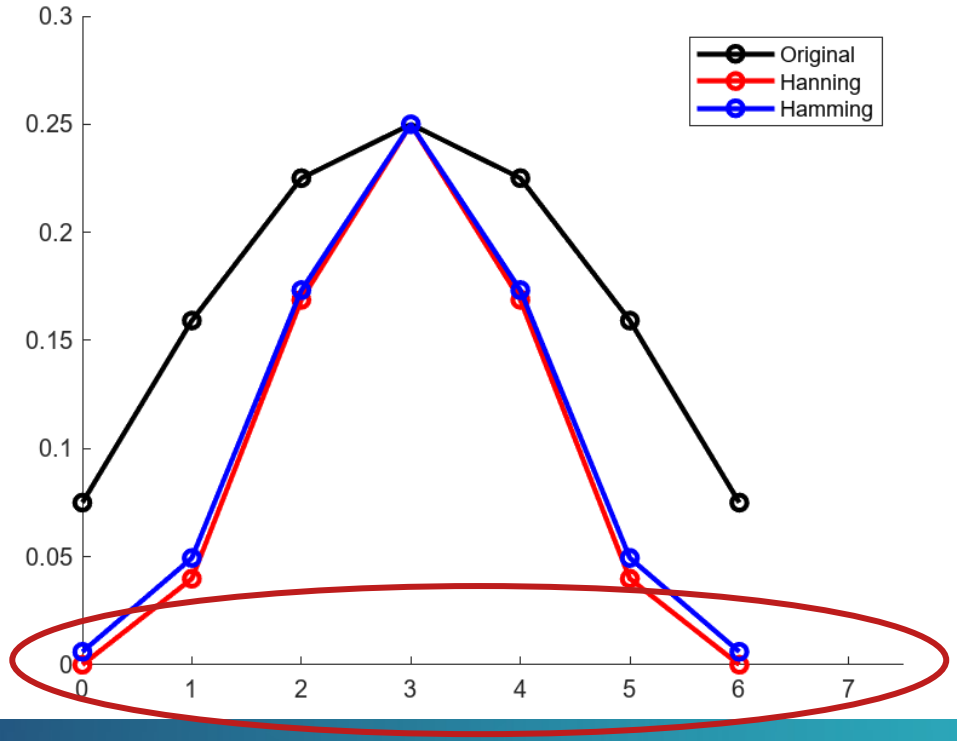


FIR Filter – Window Method

- Example: Given the filter impulse response

3. Plot $h[n], h_{han}[n], h_{ham}[n]$

Now we have a causal FIR filter



FIR Filter – Window Method (MATLAB)

- MATLAB has built-in functions to compute the window functions:
  - Rectangular window
    → rectwin(M)
  - Hanning window
    → hann(M)
  - Hamming window
    → hamming(M)

```
w_rec=transpose(rectwin(7))
```

```
w_rec = 1×7
     1     1     1     1     1     1     1
```

```
w_han=transpose(hann(7))
```

```
w_han = 1×7
         0    0.2500    0.7500    1.0000    0.7500    0.2500         0
```

```
w_ham=transpose(hamming(7))
```

```
w_ham = 1×7
    0.0800    0.3100    0.7700    1.0000    0.7700    0.3100    0.0800
```

Same results to our previous calculations

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FIR Filter Example – Moving Average Filter

- The **moving average filter** is the most common filter in DSP that *smooth out a signal by averaging over a window of data points*

- In its simplest form, a moving average filter of length M takes the average of every M consecutive samples of the waveform.

- A moving average filter can be defined by the following difference equation:

$$y[n] = \frac{1}{C} \sum_{j=-N}^{M-N-1} \alpha_j x[n+j]$$

- ✓ M is the window length
- ✓ $N < M$ specifies the number of past inputs to be considered
- ✓ $C = \sum \alpha_j$ is the normalizing constant

$$y[n] = \frac{\alpha_{-N} x[n-N] + \cdots + \alpha_0 x[n] + \cdots + \alpha_{M-N-1} x[n+M-N-1]}{\alpha_{-N} + \cdots + \alpha_0 + \cdots + \alpha_{M-N-1}}$$

FIR Filter Example – Moving Average Filter

- The following filters are all moving average filters:

1. $y[n] = \frac{x[n]+x[n+1]+x[n+2]}{3}$

2. $y[n] = \frac{x[n-2]+x[n-1]+x[n]}{3}$

One-sided moving average filter
The most simplest and common form

3. $y[n] = \frac{x[n-1]+x[n]+x[n+1]}{3}$

Symmetric moving average filter

4. $y[n] = \frac{x[n-1]+2x[n]+x[n+1]}{4}$

5. $y[n] = \frac{1}{5}(3x[n] + x[n-1] + x[n-2])$

Weighted moving average filter

FIR Filter Example – Moving Average Filter

- Write out the impulse response of the following filters:

1. $y[n] = \frac{x[n]+x[n+1]+x[n+2]}{3}$

2. $y[n] = \frac{x[n-2]+x[n-1]+x[n]}{3}$

3. $y[n] = \frac{x[n-1]+x[n]+x[n+1]}{3}$

4. $y[n] = \frac{x[n-1]+2x[n]+x[n+1]}{4}$

5. $y[n] = \frac{1}{5}(3x[n] + x[n-1] + x[n-2])$

FIR Filter Example – Moving Average Filter

- Write out the impulse response of the following filters:

1. $y[n] = \frac{x[n]+x[n+1]+x[n+2]}{3}$
2. $y[n] = \frac{x[n-2]+x[n-1]+x[n]}{3}$
3. $y[n] = \frac{x[n-1]+x[n]+x[n+1]}{3}$
4. $y[n] = \frac{x[n-1]+2x[n]+x[n+1]}{4}$
5. $y[n] = \frac{1}{5}(3x[n] + x[n-1] + x[n-2])$

1. $h[n] = \begin{cases} \frac{1}{3} & n = 0, -1, -2 \\ 0 & otherwise \end{cases}$
2. $h[n] = \begin{cases} \frac{1}{3} & n = 0,1,2 \\ 0 & otherwise \end{cases}$
3. $h[n] = \begin{cases} \frac{1}{3} & n = -1,0,1 \\ 0 & otherwise \end{cases}$

Rectangular impulse response with an area of 1

FIR Filter Example – Moving Average Filter

- Write out the impulse response of the following filters:

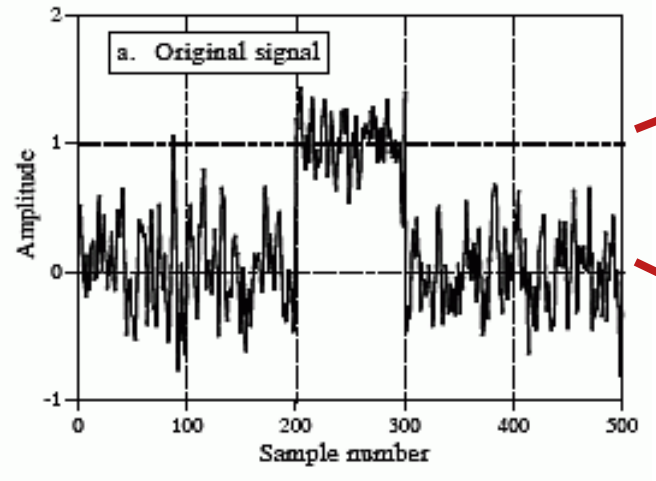
1. $y[n] = \frac{x[n]+x[n+1]+x[n+2]}{3}$
2. $y[n] = \frac{x[n-2]+x[n-1]+x[n]}{3}$
3. $y[n] = \frac{x[n-1]+x[n]+x[n+1]}{3}$
4. $y[n] = \frac{x[n-1]+2x[n]+x[n+1]}{4}$
5. $y[n] = \frac{1}{5}(3x[n] + x[n-1] + x[n-2])$

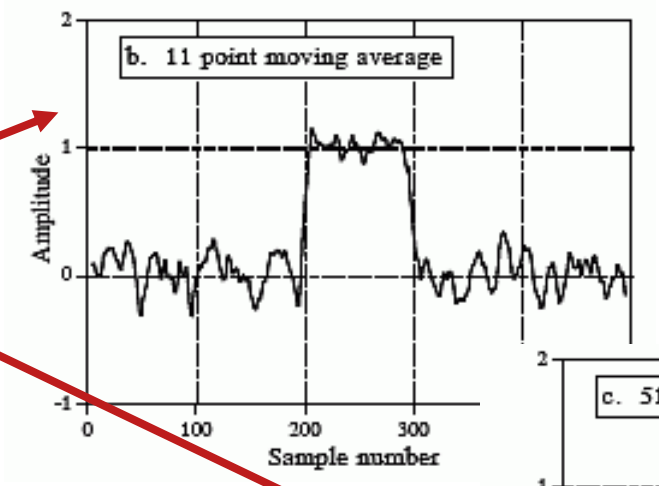
4. $h[n] = \begin{cases} \frac{1}{4} & n = -1,1 \\ \frac{1}{2} & n = 0 \\ 0 & otherwise \end{cases}$

5. $h[n] = \begin{cases} \frac{1}{5} & n = 1,2 \\ \frac{3}{5} & n = 0 \\ 0 & otherwise \end{cases}$

For moving average filters, the area under the impulse response is always 1.

FIR Filter Example – Moving Average Filter





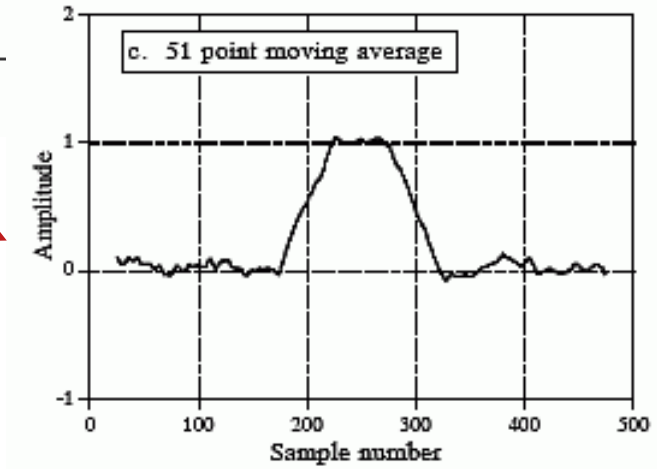


Fig (a). A rectangular pulse buried in random noise

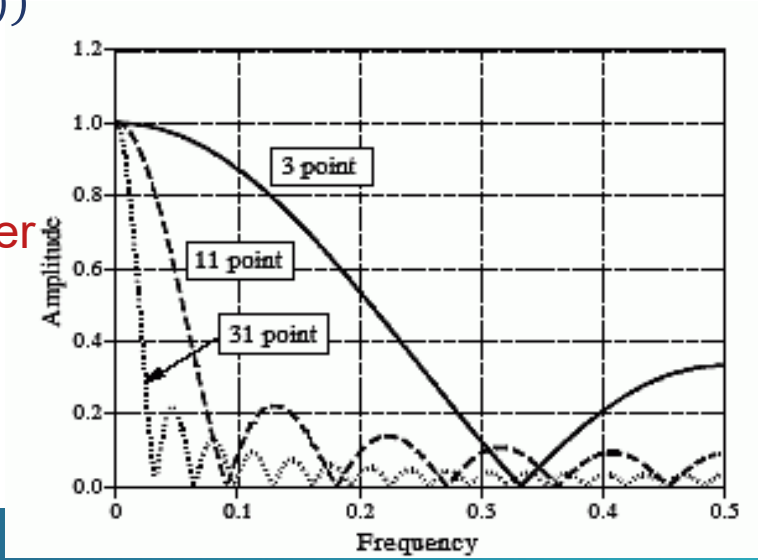
- the smoothing action of the moving average filter decreases the amplitude of the random noise (good), but also reduces the sharpness of the edges (bad)

FIR Filter Example – Moving Average Filter

- The frequency response of the moving average filter is described by the Fourier transform of the rectangular impulse response
- The amplitude response of an M point moving average filter is

$$|H(\omega)| = \frac{\sin(\omega M/2)}{M(\sin(\omega/2))}$$

- It is characterised by a slow roll-off and poor stop-band attenuation
- Hence, the moving average is a poor low-pass filter



FIR Filter Example – Moving Average Filter

- The moving average is the most common filter in DSP, mainly because it is the easiest digital filter to understand and use.
- In spite of its simplicity, the moving average filter is optimal for a common task: reducing random noise while retaining a sharp step response.
- This makes it the premier filter for time domain encoded signals.
- However, the moving average is the worst filter for frequency domain encoded signals, with little ability to separate one band of frequencies from another.

Lecture summary

- FIR filters is defined with a difference equation that does not contain past outputs
- In terms of the general equation

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}},$$

We define it to be a FIR filter if $a_k = 0, \forall k > 1$.

- FIR filters contains only zeros and the filter is non-recursive.
- All FIR filters are stable

Lecture summary

- FIR filters can be designed by multiplying the infinite impulse response with a certain window function
- There are three common window functions
  1. Rectangular window → abrupt truncation
  2. Hanning window → smoothing
  3. Hamming window → eliminate high frequency noises

Lecture summary

- A common FIR filter is the moving average filter
- Moving average filter is great for smoothing in the time-domain
- It has a *sharp step response* but a *poor frequency response*
- Hence, it cannot be used for signal processing in the frequency domain

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